

Nithin Kumar Reddy

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SUMMARY

CS undergrad building production-oriented AI systems: multi-agent orchestration, cloud data pipelines (Azure/Databricks), and autonomous developer tooling (AST, LSP, RAG). Backend + systems focus with FastAPI, distributed ETL, real-time monitoring, and security-oriented open source (Prowler/AWS). Seeking software engineering / platform / AI-infra internships where I can ship reliability, automation, and developer-experience improvements.

EDUCATION

PES University

B.Tech, Computer Science and Engineering

Coursework: DSA, OOAD, DBMS, OS, Networks, Compilers, Cloud Computing, ML, Big Data, Data Analysis

Aug 2023 -- May 2027

Bengaluru, Karnataka

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL, Bash

Backend & APIs: FastAPI, Node.js, Express, REST, WebSockets, JWT/OAuth, async services

Cloud & Data: Azure (ADF, Functions, ADLS Gen2), Databricks, Spark, Kafka, AWS, Docker, K8s, Linux, Git

AI/ML: Multi-agent systems, RAG, LangChain, HF Transformers, scikit-learn, TensorFlow, OpenCV, Pandas/NumPy

Frontend & Tooling: React, Playwright, FFmpeg, HTML/CSS

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Firebase, Supabase

Enterprise: SAP S/4HANA Sales & Distribution (SD)

EXPERIENCE

Center for Information Security, Forensics and Cyber Resilience (IFSCR)

Jun 2025 -- Aug 2025

Summer Research Intern -- Digital Image Forensics

PES University

- Built an end-to-end image provenance pipeline that identifies the original source of manipulated/reposted images using feature matching and temporal graph analysis.
- Modeled dissemination as a time-aware graph over users/posts and traced likely origin paths under reuploads, crops, and reposts.
- Implemented SIFT-based robust matching resilient to scale/crop/rotation; combined similarity + graph signals into confidence-ranked source prediction.
- Delivered a research prototype with reproducible evaluation for multi-hop image reuse forensics scenarios.

PROJECTS

Open Source -- Prowler (prowler-cloud/prowler)

2025 -- 2026

Python, AWS, cloud security compliance | <https://github.com/prowler-cloud/prowler>

- External open-source contributions to Prowler (widely used cloud security platform), not personal-only repos.
- PR #12076 (<https://github.com/prowler-cloud/prowler/pull/12076>): EC2 check for instances stopped longer than a configurable threshold.
- PR #12118 (<https://github.com/prowler-cloud/prowler/pull/12118>): SageMaker endpoint-config KMS encryption check + service model/tests.
- Live portfolio with contribution summary: <https://nithin078.github.io>

Aegis Engineer

2025 -- 2026

Python, multi-agent systems, AST, LSP, semantic search, GitHub API

- Autonomous software-engineering platform that ingests a repo, understands structure via AST/LSP + semantic search, and proposes production-ready PR-style fixes for GitHub issues.
- Multi-agent orchestration (repo intelligence, planning, code edit, verification) for grounded, reviewable diffs -- not free-form chat.
- Repo + demo: <https://github.com/Nithin078/Aegis-Engineer> | Live: <https://nithin078.github.io>

Agentic Orchestration for Big Data Pipelines

Jan 2025 -- May 2025

Python, FastAPI, Azure Data Factory, Databricks, React, scikit-learn

- 7-agent system (Planner, Resource, Performance Prediction, Monitor, Assurance, Executor, Central Manager) orchestrating ETL on ADF + Databricks.
- HistGradientBoosting models for automated compute sizing; ~30% over-provisioning reduction in evaluation runs.
- Real-time monitoring (WebSockets), LLM anomaly detection, mid-run reallocation; ~25% fewer simulated pipeline failures; dual-validation Assurance Agent + React analytics dashboard.
- Repo: <https://github.com/Nithin078/Agentic-AI-Driven-Dynamic-Orchestration-of-Big-Data-Pipelines> | Live: <https://nithin078.github.io>

DemoGen

Mar 2026 -- Apr 2026

Python, FastAPI, React, Playwright, FFmpeg, TTS

- AI platform that turns product walkthroughs into narrated demo videos: Playwright automation + screen capture + TTS + FFmpeg assembly.
- FastAPI + React control plane for multi-step UI flows, status tracking, and downloadable outputs.

- Repo: <https://github.com/Nithin078/DemoGen> | Live demo walkthrough: <https://nithin078.github.io/#demo>

ACHIEVEMENTS & CERTIFICATIONS

- Top 4 -- FASTATHON, SRM University All-India Hackathon
- Top 4 -- Space Race Hackathon, Equinox Club, PES University
- Participant -- Adobe India Hackathon
- Oracle Cloud Infrastructure -- AI Foundations Associate
- HackerRank -- Problem Solving (Intermediate)